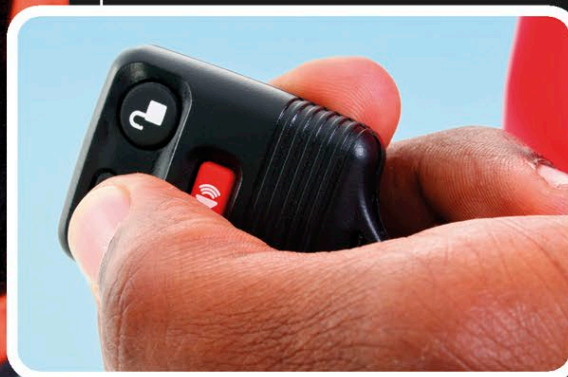


EMERGENCY RESCUE

RADIO COMMUNICATION

Long-distance communication was slow and difficult for centuries before the discovery of radio waves made wireless communication possible. Radio waves are a type of electromagnetic radiation (see page 142) that can be used to send signals over large distances at great speed. Many modern communication methods involve a device called a transceiver, which contains antennas that can both send and receive radio waves, acting as a transmitter as well as a receiver. Radio communication is used in mobile phones, radio, radar, Wi-Fi, and other applications.



GOING KEYLESS

A smart key fob is a small electronic device that allows access to a vehicle without using a regular key. When triggered, a transmitter sends a radio signal to a receiver that locks or unlocks the door.

Many objects in space, such as the Sun and other large stars, emit radio waves.

RADIO SIGNALS

Radio waves have the longest wavelengths in the electromagnetic spectrum, which enable them to travel far. They are generated by a transmitter using changing electric currents. By adding a signal to a radio wave, information can be encoded in the wave, which is then radiated outward from antennas. A receiver works like a transmitter in reverse, receiving the waves and turning them back into electric currents.

